

Building Technology (AR 246) Past Questions Analysis (2079 - 2082)

Unit 1: Introduction - 8.0% of Total Marks (Syllabus Hours: 3)

A. Short Answer Questions (SAQs)

2082 Q1 (5 Marks):

Explain the functional planning of buildings. Discuss the principles of site selection, set-back, floor space index (FSI), open space, and orientation of buildings with suitable examples.

2081 Q1 (8 Marks):

What do you understand by building orientation? Mention the principles of site selection. **(2+6 Marks)**

2081 Mid-Term Q6 (8 Marks):

Why set-back is necessary for building construction? Explain with necessary sketches. **[SKETCH REQUIRED: Sketch showing the property line, setback line, building envelope, and the clear margin of open space left around the building structure.]**

2080 Q1 (8 Marks):

What are the various components of the building and their functional requirement in the building?

2079 Q1 (8 Marks):

Explain the process of setting layout plan of a building with necessary sketches. **[SKETCH REQUIRED: Draw a schematic diagram of a grid/centerline setting-out plan, showing profiles (batter boards), pegging, string lines, and the method of checking the 3-4-5 right-angle rule.]**

B. Long Answer Questions (LAQs)

Historically, Unit 1 questions are structured as 8-mark questions, which fall under Group A (Short Answer) in the 100-mark papers or Group A (5-mark) in the 60-mark paper. No 15-mark or 10-mark long questions have been isolated solely on Unit 1 introduction.

C. Trend Analysis & Focal Areas

- ***Question Types & Styles:*** Focuses entirely on the fundamental planning principles of building design. Expect descriptive questions on building orientation, site selection criteria, and zoning definitions.

- **Direct Repetitions:** Orientation and site selection principles are highly repeated (2082 Q1, 2081 Q1). Set-back requirements and setting-out layout plan techniques are crucial (2081 Mid-Term, 2079 Q1).
- **Key High-Priority Topics:**
 - Principles of Site Selection: Orientation, soil conditions, wind, and water source accessibility.
 - Zoning & Regulations: Building Bye-laws (Setback, Floor Space Index (FSI), Open Space percentages).
 - Setting Out: The step-by-step practical process of laying out foundations on a site (profile boards and ropes).

Unit 2: Foundation - 14.7% of Total Marks (Syllabus Hours: 4)

A. Short Answer Questions (SAQs)

2082 Q2 (5 Marks):

a) It is noticed that various sites are confronted with the problem of soil failure during excavation. How can you solve such issues? What different methods can be employed to enhance the bearing capacity of weak soils? **(1.5+1.5 Marks)** b) Discuss about the functions of geotextiles. **(2 Marks)**

2081 Q2 (8 Marks):

List out the types and functions of geo-textile. State the major causes of failure of foundation. **(4+4 Marks)**

2081 Mid-Term Q1 (8 Marks):

In your locality the bearing capacity of soil is low, in this situation which type of foundation do you apply for multistory building and why? What are the common problems with existing foundation?

2080 Q2 (8 Marks):

Explain about various types of foundations with sketches. What are the various techniques of dewatering of foundation trenches? **(4+4 Marks)** **[SKETCH REQUIRED: Sketch basic shallow foundation types (strip footing, pad/column footing, stepped footing, and raft/mat foundation slab) and a simple well-point or sump-pump dewatering system.]**

2080 Q10b (5 Marks - Short Note):

Geo textile.

2079 Q2 (8 Marks):

What are the remedial precautions to be undertaken to prevent unequal settlement of the existing foundation? What type of foundation will you recommend for the foundation on black cotton soils? Give your justification.

2079 Q5 (8 Marks):

Due to natural disaster one footing of a building is completely damaged. In this situation which type of solution do you apply for this building & how it can be implemented? Also write merits and demerits of the proposed solution.

B. Long Answer Questions (LAQs)

2080 Q8a (3 Marks part of Q8):

What are the various causes of foundation settlement? *(Combined with underpinning in 2080 Q8, total 8 marks).*

2079 Q7a (8 Marks part of Q7):

Differentiate pile foundation, pier foundation & caisson foundations. *(Combined with stone masonry in 2079 Q7, total 15 marks).*

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** *This is a heavy, core unit of civil engineering. Questions balance structural decisions (choosing pile vs. shallow bases) with geotechnical solutions (dewatering, soil improvement, and geotextiles).*
- **Direct Repetitions:** *Geotextiles functions and applications appear as a core question or short note in 2082, 2081, and 2080. Dewatering methods and the causes of foundation settlement/failure are highly repeated.*
- **Key High-Priority Topics:**
 - *Black Cotton Soil Foundations: Remedial actions and choices (e.g., under-reamed piles, raft foundations).*
 - *Dewatering Techniques: Sump-pumping, deep well points, and vacuum methods.*
 - *Geotextiles: Separation, filtration, drainage, reinforcement, and protection functions.*
 - *Pile vs. Pier vs. Caisson: Precise structural definitions and soil suitability.*

Unit 3: Walls and Damp Proofing - 12.8% of Total Marks (Syllabus Hours: 5)

A. Short Answer Questions (SAQs)

2082 Q3 (5 Marks):

A residential building is located in a high-rainfall area with high humidity. Which type of wall (solid brick wall, cavity wall, or curtain wall) would you recommend and why?

2081 Q3 (8 Marks):

Mention the purpose and methods for damp proofing.

2081 Mid-Term Q2 (8 Marks):

In hilly area generally which type of bond is applied for building construction and why? Also write merit and demerit of that bond.

2080 Q3 (8 Marks):

What are the requirements for a good partition wall? Draw a typical section of an outer wall showing all the details required for construction of wall. **(4+4 Marks) [SKETCH REQUIRED: Draw a detailed cross-section of a standard 230mm brick outer wall from the concrete footing through the DPC level up to the plinth and superstructure, showing plastering layers.]**

2080 Q10a (5 Marks - Short Note):

Damp Proofing in Wall.

B. Long Answer Questions (LAQs)

2082 Q7b (2 Marks part of Q7):

During field investigation it is seen that some of the buildings face seepage problems. Suggest different remedial measures to solve such issues.

2082 Q7c (3 Marks part of Q7):

Why termite treating is inevitable in building? Illustrate about the methods of termite treating.

2079 Q7b (7 Marks part of Q7):

Define stone masonry with its types. Draw typical sketches to illustrate them. **[SKETCH REQUIRED: Sketch Rubble Masonry (Uncoursed, Coursed) and Ashlar Masonry (Ashlar Fine, Ashlar Rough/Champlered) showing stone alignments.]**

2079 Q8a (8 Marks part of Q8):

Define the term bond. What are the types of bond used in brick masonry? Explain any four of them with neat sketches. **[SKETCH REQUIRED: Sketch plan views of alternative courses for English Bond and Flemish Bond (Header-Stretcher alternations) for a 1-brick and 1.5-brick thick wall.]**

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Heavy emphasis on construction craftsmanship (brick/stone masonry bonds) and protection systems (Damp Proof Course - DPC, termite prevention, seepage remedies).
- **Direct Repetitions:** Brick bonds (English and Flemish) and stone masonry types are highly repeated. Damp-proofing methods and DPC materials are guaranteed points of evaluation across the papers.
- **Key High-Priority Topics:**
 - *Bonds in Masonry:* English Bond vs. Flemish Bond (sketches of headers, stretchers, queen closers).
 - *Damp-proofing (DPC):* Causes of dampness (capillary action), materials used (bitumen, cement concrete with water-proofers), and installation methods at plinth level.
 - *Cavity Walls:* Thermal and moisture insulation benefits in high-humidity/rainfall areas.

Unit 4: Stair, Lifts and Escalator - 8.1% of Total Marks (Syllabus Hours: 2)

A. Short Answer Questions (SAQs)

2082 Q4 (5 Marks):

Design a Proper Layout of a RCC stair for a hospital from the following data: Inside dimension of staircase: 6.00m x 4.50m, Height of Floor: 3.75m, Thickness of floor slab: 150mm, Width of stair (flight): 1.50m. **[SKETCH REQUIRED: Draw a plan and sectional elevation of the designed dog-legged stair, labeling Rise, Tread, Flight, Landing, Waist slab, Headroom, and Handrail height.]**

2081 Q4 (8 Marks):

Design a proper layout of a RCC stair for a hospital from the following data: Inside dimension of staircase - 6.00 x 4.50, Height of floor - 3.75, Thickness of floor slab - 150mm, Width of stair - 1.50m. Assume the necessary data and draw neat sketch to support the design. **[REPEATED:**

Identical values to 2082 Q4 and 2081 Mid-Term Q3. SKETCH REQUIRED: Plan and sectional elevation showing the layout, step sizing, and waist slab thickness.]

2081 Mid-Term Q3 (8 Marks):

Design a Proper Layout of a RCC stair for a hospital from the following data: Inside dimension of staircase: 6.00m x 4.50m, Height of Floor: 3.75m, Thickness of floor slab: 150mm, Width of stair (flight): 1.50m. Assume the necessary data and draw a neat sketch to support the design.

[REPEATED: Identical values to 2082 Q4. SKETCH REQUIRED: Dog-legged stair layout showing flights and landings.]

2080 Q4 (8 Marks):

Design and draw plan of a 1.2 m wide doglegged stair for a residential building with 3m floor height. **[SKETCH REQUIRED: Plan showing staircase dimensions, steps (usually 10 risers per flight), mid-landing, and flight width.]**

2079 Q3 (8 Marks):

A stair hall measure 4m x 4m with floor height 3.3m. Design a suitable stair for the building with necessary sketches. **[SKETCH REQUIRED: Plan view of the designed stair (either dog-legged or open-well depending on calculations) detailing treads, risers, and landings.]**

B. Long Answer Questions (LAQs)

No Unit 4 questions have been asked in the Group B 15-mark/10-mark sections. They are exclusively asked as design-based SAQs (Group A).

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** *This chapter features **guaranteed design numericals**. You will always be given floor height and staircase room dimensions and asked to calculate the number of risers, treads, flight dimensions, landings, and draw a layout.*
- **Direct Repetitions:** **THE HOSPITAL STAIRCASE (6.0m x 4.5m, 3.75m floor height, 1.5m stair width) WAS ASKED IDENTICALLY THREE TIMES: 2082, 2081, AND 2081 MID-TERM!** Master this exact design!
- **Design Guide Template for Hospital staircase:**
 1. Total Height: 3.75 m = 3750 mm. Height of one flight (mid-landing) = 1875 mm.
 2. Riser (R): Standard for public/hospital = 150 mm. Number of risers = $3750 / 150 = 25$ (i.e., 12 in one flight, 13 in the other, or use 150 mm for 25 risers with asymmetric landings).
 3. Tread (T): Standard for public = 300 mm. Number of treads per flight = $R - 1 = 12 - 1 = 11$ steps.

4. *Total Run: $11 \times 300 \text{ mm} = 3300 \text{ mm} = 3.3 \text{ m}$.*
 5. *Staircase Room Check: Total length available is 6.0 m. Run takes 3.3 m. Leftover space for landings is $6.0 - 3.3 = 2.7 \text{ m}$. This allows for a very comfortable starting landing of 1.35 m and mid-landing of 1.35 m (both exceeding the required 1.5 m flight width rule if layout adjustments are optimized).*
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Unit 5: Floors and Ceilings - 5.3% of Total Marks (Syllabus Hours: 2)

A. Short Answer Questions (SAQs)

2082 Q5 (5 Marks):

Compare solid and suspended floors. Suggest suitable floor and ceiling finishes for a multi-storey residential building, justifying your choices based on durability, cost, and functionality.

2081 Mid-Term Q5 (8 Marks):

Explain the detail about floor and ceiling with necessary sketches. **[SKETCH REQUIRED: Sketch a cross-section of a Ground Solid Floor (rammed earth, sand filling, dry brick flat soling, cement concrete, and terrazzo/mosaic finish) and an Upper Suspended Timber or RCC floor.]**

2079 Q4b (4 Marks part of Q4):

Show flooring details of a ground floor that has a connection with external wall and floor finish with mosaic tiles. **[SKETCH REQUIRED: Cross-sectional diagram of the junction of the 230mm brick wall with the solid ground floor, clearly illustrating the location of the damp-proof course (DPC), brick soling, concrete bed, and the mosaic tile topping.]**

B. Long Answer Questions (LAQs)

2081 Q11c (5 Marks - Short Note):

Solid and suspended floor.

2080 Q10c (5 Marks - Short Note):

Solid and suspended Floor.

C. Trend Analysis & Focal Areas

- ***Question Types & Styles:*** Focused on contrasting ground-supported (solid) floors with suspended (upper floor) structural configurations, alongside the details of plastering and finishes.

- **Direct Repetitions:** The contrast between **solid and suspended floors** is a constant repeat (2082 Q5, 2081 Q11c, 2080 Q10c). Floor finishing details (like mosaic/terrazzo soling layers) are highly tested.
 - **Key High-Priority Topics:**
 - **Solid Floors (Ground):** Layer-by-layer composition from earth subgrade to finish tiles.
 - **Suspended Floors:** Steel truss, timber joints, or RCC slab configurations for upper floors.
 - **Finishes:** Mosaic tiles, cement screed, parquet, and vinyl finishes with selection criteria.
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Unit 6: Roofs - 7.8% of Total Marks (Syllabus Hours: 3)

A. Short Answer Questions (SAQs)

2082 Q10a (5 Marks):

A residential building is located in a region with heavy rainfall and high wind. Recommend a suitable roof type, truss material, and roof covering, justifying your choices in terms of drainage, load-bearing capacity, durability, and cost.

2081 Q5 (8 Marks):

Mention the requirements of a good roof. Explain the king post truss and Queen post truss roof with their necessary sketches. **(2+3+3 Marks) [SKETCH REQUIRED: Detailed line sketches of a King Post Truss (central post, principal rafters, struts, tie beam, purlins) and a Queen Post Truss (two vertical queen posts, straining beam, straining sill, struts, rafters, tie beam).]**

2081 Mid-Term Q4 (8 Marks):

Differentiate between king post and queen post roof truss with necessary sketches.

[REPEATED: Similar to 2081 Q5 and 2079 Q10a. SKETCH REQUIRED: Comparative drawings of the internal member arrangement of King Post (span 5-9m) vs. Queen Post (span 9-14m) timber trusses.]

2080 Q5 (8 Marks):

What is roof used for? Sketch out different types of roof and show their parts. **(3+5 Marks) [SKETCH REQUIRED: Simple line sketches of a Gable roof, Hip roof, Lean-to roof, Mansard roof, and Flat roof slab with parapet walls.]**

B. Long Answer Questions (LAQs)

2079 Q10a (7 Marks part of Q10):

Differentiate between king post and queen post roof truss with necessary sketches.

[REPEATED: Identical to 2081 Mid-Term Q4. SKETCH REQUIRED: Structural member layouts detailing connections.]

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** *This chapter is **extremely visual**. King Post and Queen Post timber trusses are the ultimate staples here and must be practiced until you can draw them rapidly from memory.*
- **Direct Repetitions:** *The difference between King Post and Queen Post roof trusses was asked in 2081, 2081 Mid-Term, and 2079!*
- **Key High-Priority Topics:**
 - **King Post Truss (Span: 5 to 9 meters):** *Single central vertical post under tension, two rafters under compression, two struts.*
 - **Queen Post Truss (Span: 9 to 14 meters):** *Two vertical posts under tension connected by a straining beam and a straining sill.*
 - **Roof Coverings:** *GI sheets, CGI sheets, clay tiles, and slate, with suitability for sloping roofs.*

Unit 7: Doors, Windows, Ventilations and Skylights Openings - 4.7% of Total Marks (Syllabus Hours: 3)

A. Short Answer Questions (SAQs)

2082 Q10b (5 Marks):

A building is located in a region with extreme weather conditions and high humidity. Recommend suitable materials, sizes, and design features for doors and windows to ensure durability, energy efficiency, and comfort. Justify your choices with relevant examples and sketches. **[SKETCH REQUIRED: Draw elevations of a fully paneled timber door and a double-glazed aluminum casement window, highlighting weather-stripping, sills, and double glazing.]**

2081 Q6 (8 Marks):

Explain the various openings used in a residential building along with their functions and types.

2079 Q4a (4 Marks part of Q4):

In your opinion which type of door is suitable for residential building & why? (Combined with floor details in 2079 Q4, total 8 marks).

B. Long Answer Questions (LAQs)

Exclusively examined as short-answer conceptual/descriptive parts under Group A.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** *Focuses on functional design (energy efficiency, climate response) and selection of materials (timber, aluminum, UPVC).*
 - **Key High-Priority Topics:**
 - **Door Types:** *Fully paneled doors (best for aesthetic/durability in residential), glazed doors, flush doors.*
 - **Energy-Efficient Windows:** *Double-glazing (argon gas filling), low-E coating, thermal break frames.*
 - **Components:** *Sash, frame, sill, lintel, and mullion.*
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Unit 8: Joints and Fire Place - 6.9% of Total Marks (Syllabus Hours: 3)

A. Short Answer Questions (SAQs)

2081 Q11b (5 Marks - Short Note):

Joints in Building.

B. Long Answer Questions (LAQs)

2082 Q6a (5 Marks part of Q6):

A long multi-storey building is exposed to extreme temperature changes and requires a residential fireplace in the living area. Recommend suitable types of joints for the building and explain the design of the fireplace, including flue, stack, insulation, and materials used. Justify your choices. (Combined with temporary construction in 2082 Q6, total 10 marks).

2080 Q8b (7 Marks part of Q8):

Why joints are necessary in the building construction? Describe in detail about the types of joints in structure and their uses. **(2+5 Marks)** (Combined with underpinning in 2080 Q8, total 15 marks).

2079 Q10b (8 Marks part of Q10):

Define joints with their different types. Where and how construction joints are provided for RCC beam, column and slab? (Combined with King/Queen post in 2079 Q10, total 15 marks).

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Focuses on structural joints (construction joints, expansion joints) and the fireplace draft and flue mechanics.
- **Direct Repetitions:** The types, necessity, and methods of construction for **expansion and construction joints** are highly repeated across 2082, 2081, 2080, and 2079.
- **Key High-Priority Topics:**
 - **Expansion Joints vs. Construction Joints:** Expansion joints allow thermal movement (full gap through structure). Construction joints are stopping points in concreting (require rough surfaces and dowel bars).
 - **RCC Construction Joints Placement:**
 - **Beams & Slabs:** Placed at mid-span or within the middle third of the span where shear force is minimum.
 - **Columns:** Placed horizontally just below the beam-column junction.
 - **Fireplace Design Components:** Hearth, firebox, throat, damper, smoke chamber, flue liner, chimney stack.

Unit 9: Temporary Construction - 9.2% of Total Marks (Syllabus Hours: 4)

A. Short Answer Questions (SAQs)

No SAQs isolated in this unit. Questions are consistently asked as heavy, high-scoring Group B LAQs.

B. Long Answer Questions (LAQs)

2082 Q6b (5 Marks part of Q6):

A multi-storey building is under construction with RCC beams and slabs, and heavy masonry walls. Recommend suitable types of scaffolding, shoring, underpinning, and formwork to be used during building construction, explaining the materials, techniques, and safety considerations for each. (Combined with joints/fireplace in 2082 Q6, total 10 marks).

2081 Q7 (15 Marks):

In what situations scaffolding, shoring and under-pinning are required? Explain with neat sketches. (5+5+5 Marks) **[SKETCH REQUIRED: Sketch a Bricklayer's Single Scaffolding (ledger, standards, putlogs, platform), a Raking Shore (rakers, cleat, needle, sole plate), and a Pit/Pile Underpinning setup.]**

2080 Q7a (8 Marks part of Q7):

Define Shoring. What are the objectives of shoring? Describe the types of shoring with necessary sketches. (1+3+4 Marks) **[SKETCH REQUIRED: Sketch a Raking Shore, Flying Shore (between two adjacent walls), and Dead Shore (vertical needles supporting a wall from below) with detailed labels.]**

2080 Q8a (5 Marks part of Q8):

Define underpinning and explain different methods of underpinning with neat sketches. (Combined with foundation settlement in 2080 Q8, total 8 marks). **[SKETCH REQUIRED: Sketch Pit Underpinning showing sequence of excavation and concrete packing under an existing wall footing.]**

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** This is a *critically high-scoring chapter*. The trio of **Shoring, Underpinning, and Scaffolding** is tested thoroughly, requiring detailed definitions, operational circumstances (when to use), and neat sketches.
- **Direct Repetitions:** **Shoring, Underpinning, and Scaffolding** are asked as a combined 15-mark package or major sub-questions in every single paper (2082, 2081, 2080).
- **Key High-Priority Topics:**
 - **Shoring:** Raking Shores (lateral support), Flying Shores (cross-supports between buildings), and Dead Shores (vertical support during foundation repair).
 - **Underpinning:** Methods (Pit method, Pile method) used to strengthen existing weak foundations.
 - **Scaffolding:** Single (bricklayers) vs. Double (masons) scaffolding, and tubular steel frame scaffolding.

Unit 10: Protection Works of Building - 16.1% of Total Marks (Syllabus Hours: 3)

A. Short Answer Questions (SAQs)

2082 Q7a (5 Marks):

State the need of retrofitting. What are the factors that should be considered in order to decide whether to retrofit a building or not?

2079 Q6 (8 Marks):

What do you mean by shrinkage cracks? Explain the factors that are adopted for preventing shrinkage cracks.

B. Long Answer Questions (LAQs)

2081 Q9 (15 Marks):

Explain the causes and prevention of cracks in walls and plasters. What do you mean by non-destructive tests? Describe any two of them. **(8+1+6 Marks)**

2080 Q9a (8 Marks part of Q9):

What are the various non-destructive tests in the building structure? What may be the causes of occurrence of cracks in the building and what are the remedial measures to cracks? **(3+5 Marks)**
(Combined with house wiring in 2080 Q9, total 15 marks).

2079 Q8b (7 Marks part of Q8):

State need of retrofitting. What are the factors that should be considered in order to decide whether to retrofit or not? **[REPEATED: Identical to 2082 Q7a.]**

2079 Q11b (8 Marks part of Q11):

Why non-destructive test is used in building? Which method do you apply for non-destructive test on structures? (Combined with earthing/CCTV in 2079 Q11, total 15 marks).

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Highly practical and maintenance-focused. This unit consistently questions why structures degrade (cracking, seismic vulnerability) and how engineers test (Non-Destructive Testing) and strengthen them (retrofitting).
- **Direct Repetitions:**
 1. The exact question on "Need of retrofitting and factors to decide" repeated in 2082 Q7a and 2079 Q8b.
 2. NDT (Non-Destructive Testing) methods and applications repeated in 2081 Q9, 2080 Q9a, and 2079 Q11b.

3. *Causes and prevention of cracks in building elements repeated in 2081, 2080, and 2079.*

- **Key High-Priority Topics:**

- **Retrofitting Factors:** Structural age, seismic zone, building change-of-use, extent of current damage, economic feasibility.
- **NDT Methods:** Rebound Hammer Test (surface hardness/compressive strength estimation) and Ultrasonic Pulse Velocity (UPV) Test (internal concrete quality/cracks detection).
- **Cracks in Buildings:** Structural (load-induced, foundation settlement) vs. Non-structural (shrinkage, thermal, chemical action).

Unit 11: External and Internal Wall Finishing - 2.8% of Total Marks
(Syllabus Hours: 2)

A. Short Answer Questions (SAQs)

No SAQs isolated in this unit.

B. Long Answer Questions (LAQs)

2081 Q11a (5 Marks - Short Note part of Q11):

External wall finishing.

2080 Q10e (5 Marks - Short Note part of Q10):

Wall Finishing.

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Historically examined as a short-note topic under Group B. Focuses on the aesthetic and weather-protection layers of the building envelope.
- **Key High-Priority Topics:**
 - **Plastering & Pointing:** Composition of mortar mixes (cement-sand, lime), and finishing pointing styles (recessed, flush, ruled).
 - **Wall Cladding:** Decorative brick tiling, stone veneer siding, and ACP (Aluminum Composite Panel) cladding.

Unit 12: Water Supply and Sanitary Works - 9.4% of Total Marks

(Syllabus Hours: 3)

A. Short Answer Questions (SAQs)

Typically examined as a major 15-mark or 10-mark Group B design-and-sketch question.

B. Long Answer Questions (LAQs)

2082 Q8a (5 Marks part of Q8):

A residential building faces water logging and foul smell from drains. Suggest suitable drainage, septic tank, soak pit, and pipe layout, explaining pipe gradients and placement of traps for proper hygiene with necessary drawings. *(Combined with insulation in 2082 Q8, total 10 marks).*

[SKETCH REQUIRED: Sketch a simple building sanitary plumbing schematic, showing soil pipe, waste pipe, gully trap, floor trap (P-trap), and connection to a septic tank.]

2081 Q8 (15 Marks):

Discuss the rain water harvesting methods for domestic purpose. Draw the labeled sectional diagrams of septic tank and soak pit. **(5+5+5 Marks)** *[SKETCH REQUIRED: 1. Rainwater harvesting schematic showing roof catchments, gutters, down-take pipes, filter chamber, and storage tank. 2. Cross-sectional elevation of a Septic Tank (baffle walls, inlet/outlet pipes, sludge zone). 3. Soak Pit elevation showing brick packing and aggregate layer filters.]*

2080 Q7b (7 Marks part of Q7):

Describe a septic tank and soak pit with necessary sketches. *[REPEATED: Similar to 2081 Q8. SKETCH REQUIRED: Cross-sections showing working liquid depths and vent pipes.]*

2079 Q9b (7 Marks part of Q9):

What is septic tank? Explain the construction details of a septic tank with a neat sketch. *(Combined with wiring in 2079 Q9, total 15 marks). [REPEATED: Similar to 2080 Q7b. SKETCH REQUIRED: Labeled drawing of a two-chambered septic tank detailing the inlet, outlet, baffle wall, and manhole cover.]*

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** *This is a guaranteed high-scoring visual unit. It demands heavy practical sketches of wastewater disposal systems.*
- **Direct Repetitions:** *Septic tank and soak pit construction details and sectional drawings are asked in 2082, 2081, 2080, and 2079! This is a 100% recurring topic.*
- **Key High-Priority Topics:**

- **Septic Tank Design:** Minimum dimensions, retention period, function of baffle walls (separating scum, effluent, and sludge), and vent pipe necessity.
- **Soak Pit:** Mechanics of dispersing treated effluent into soil safely through percolation.
- **Rainwater Harvesting:** Components (collection, filtration, storage) and plumbing arrangements.

Unit 13: Thermal and Sound Insulation - 3.6% of Total Marks **(Syllabus Hours: 3)**

A. Short Answer Questions (SAQs)

2080 Q6 (8 Marks):

Explain the various measures taken for the thermal and sound insulation in a building.

B. Long Answer Questions (LAQs)

2082 Q8b (5 Marks part of Q8):

Suggest thermal and sound insulation methods for a building in a humid, noisy area. Include moisture control, insulation materials, and noise reduction techniques with justification.

(Combined with sanitary/drainage in 2082 Q8, total 10 marks).

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** Conceptual questions evaluating the physics of sound dampening, heat transfer control, and condensation mitigation.
- **Key High-Priority Topics:**
 - **Thermal Insulation Materials:** Glass wool, polyurethane foam, expanded polystyrene, cavity wall air gap.
 - **Sound Insulation Methods:** Double glazing, acoustic ceiling tiles, carpet flooring, heavy solid core doors, wall mass enhancement.
 - **Moisture & Condensation:** Prevention using vapor barriers on the warm side of insulation to avoid dampness.

Unit 14: Building Services - 18.6% of Total Marks (Syllabus Hours: 5)

A. Short Answer Questions (SAQs)

2080 Q10d (5 Marks - Short Note part of Q10):

Fire Protection in Building.

B. Long Answer Questions (LAQs)

2082 Q9a (5 Marks part of Q9):

Write the different wiring systems in common use with their advantages. *(Combined with earthing/CCTV in 2082 Q9, total 10 marks).*

2082 Q9b (5 Marks part of Q9):

What do you mean by earthing and why is it required for a building? Mention the uses of close-circuit television in buildings. *(Combined with wiring in 2082 Q9, total 10 marks).*

2081 Q10 (15 Marks):

There are several modern techniques that are being used in a Green Building to save the energy. Discuss the techniques and building services that could be used in a building to minimize its energy consumption.

2080 Q9b (7 Marks part of Q9):

Explain about the various components of the house wiring system. *(Combined with NDT/cracks in 2080 Q9, total 15 marks).*

2080 Q11 (15 Marks):

Write an article on Smart and Energy efficient Building.

2079 Q9a (8 Marks part of Q9):

Write the different wiring systems in common uses with their advantages. **[REPEATED: Identical to 2082 Q9a.]**

2079 Q11a (7 Marks part of Q11):

Why earthing is required for building? How to control the crime in public place by using close-circuit television? **[REPEATED: Identical to 2082 Q9b.]**

C. Trend Analysis & Focal Areas

- **Question Types & Styles:** This is the **highest-weighted chapter in the exam** because it spans multiple modern technical services (electrical wiring, earthing, CCTV security, fire protection, and energy-efficient green buildings).

- ***Direct Repetitions:***

1. *The wiring systems comparison repeated in 2082 Q9a and 2079 Q9a.*
2. *The combination of Earthing necessity and Public CCTV crime control repeated identically in 2082 Q9b and 2079 Q11a.*
3. *Green, Smart, and Energy-Efficient Building design repeated in 2081 Q10 and 2080 Q11.*

- ***Key High-Priority Topics:***

- ***Wiring Systems:*** *Cleat wiring, casing-capping wiring, batten wiring, and conduit wiring (surface and concealed - the safest, standard modern method).*
- ***Earthing:*** *Plate earthing and pipe earthing systems, protecting users from leakage currents.*
- ***CCTV & Security:*** *Strategic placement, monitoring, and deterrence of crime.*
- ***Green Building Services:*** *Solar lighting, natural ventilation, smart building HVAC controls, greywater reuse, and building automation.*

Subject Priority Planner & Study Guide

The following table ranks all 14 chapters from **highest return-on-investment** to lowest, based on historical exam weighting and repeat rates. Use this to focus your preparation!

Rank	Unit No.	Unit Name	Exam Priority Class	Historical Mark Percentage	Core Focus Areas to Pass/Score
1	Unit 14	Building Services	Highest	18.6%	Concealed conduit wiring, plate earthing benefits, CCTV roles, Green Building passive solar design and energy conservation.
2	Unit 10	Protection Works of Building	Highest	16.1%	Need and decision factors of retrofitting, NDT tests (Rebound Hammer, UPV), causes and prevention of wall/plaster cracks.
3	Unit 2	Foundation	Highest	14.7%	Solutions for black cotton soil, types and functions of geo-textiles, dewatering techniques, causes of settlement.
4	Unit 3	Walls and Damp proofing	High	12.8%	Flemish vs. English bonds (with plans), uncoursed/coursed rubble and ashlar stone masonry, DPC plinth installation.
5	Unit 12	Water supply and sanitary works	High	9.4%	Guaranteed sketches: Septic tank cross-section, Soak pit filter arrangement, Rainwater harvesting flow diagram.
6	Unit 9	Temporary construction	High	9.2%	Definitions, when to use, and detailed sketches of Shoring (Raking/Dead/Flying), Underpinning, and Scaffolding.
7	Unit 4	Stair, lifts and escalator	Medium	8.1%	Guaranteed numerical design: Dog-legged stair layout calculation and planning. <i>Master the hospital stair layout template!</i>
8	Unit 1	Introduction	Medium	8.0%	Building orientation, site selection criteria, setback regulations, and setting-out layout pegs.
9	Unit 6	Roofs	Medium	7.8%	Guaranteed sketches: King Post Truss (central post) vs. Queen Post Truss (straining beam), roof types.
10	Unit 8	Joints and fire place	Medium	6.9%	Expansion vs. Construction joints, placement of RCC construction joints, fireplace components (hearth, flue).

Rank	Unit No.	Unit Name	Exam Priority Class	Historical Mark Percentage	Core Focus Areas to Pass/Score
11	Unit 5	Floors and ceilings	Medium	5.3%	Solid ground floor soling/screed layering details vs. Suspended floor finishes.
12	Unit 7	Doors, windows, open openings	Low	4.7%	Suitability of paneled doors, energy-efficient window materials (aluminum with thermal breaks, double-glazing).
13	Unit 13	Thermal and sound insulation	Low	3.6%	Moisture control (vapor barriers), acoustic ceiling materials, heat-stopping foam insulation materials.
14	Unit 11	External and internal finishing	Low	2.8%	Cement plastering mortar mixes, styles of pointing, wall cladding (stone/ACP).

Prepared by Gemini Notebook for Civil Engineering Fourth Semester Exam Preparation.

Gemini Notebook can be inaccurate; please double check its responses.